

Bench A · decoding contract

All recorded assertions passed · {"pass":5,"mismatch":0,"missing":0,"unasserted":0}

Synthetic/manual expectations are independent of this decoder. Passing assertions do not establish device conformance or safe equipment operation. No hardware connection, network access, register writes or certification. Missing and unasserted fields are incomplete coverage. Float32 uses exact binary values; integer scaling uses exact decimal arithmetic.

Address and interpretation contract

Field / source note	Space / source address	PDU range	Type / order	Scale / offset / unit
temperature · Return temperature Synthetic reference: signed FFE7 = −25; independently expected −2.5 °C.	holding / 40001 (reference)	0–0	int16 / AB	0.1 / 0 / °C
counter · Cycle counter Synthetic reference: 0001 86A0 = 100000.	holding / 2 (pdu)	2–3	uint32 / ABCD	1 / 0 / cycles
counter_low · Counter · low-word alias Intentional alias of the second counter word; retain overlap warning.	holding / 4 (one)	3–3	uint16 / AB	1 / 0 / word
flow · Flow reference Synthetic IEEE-754 reference: 4148 0000 = 12.5 exactly.	holding / 4 (pdu)	4–5	float32 / ABCD	1 / 0 / L/min
status · Input status Separate input space; PDU 0 is not a collision with holding PDU 0.	input / 30001 (reference)	0–0	uint16 / AB	1 / 0 / code

Overlap review

holding PDU 3: counter, counter_low

Aliases remain visible. An overlap is not automatically invalid; consult the source map.

Recorded snapshots and independent expectations

Steady bench · holding (steady)

holding · start PDU 0 · 6 words

FFE7 0000 0001 86A0 4148 0000

Input register reference (input)

input · start PDU 0 · 1 words

0007

Case / field	Status	Raw / canonical bits	Actual engineering value	Independent expected / absolute tolerance
steady / temperature	pass	-25 FFE7	-2.5	-2.5 / 0
steady / counter	pass	100000 000186A0	100000	100000 / 0
steady / counter_low	pass	34464 86A0	34464	34464 / 0
steady / flow	pass	12.5 41480000	12.5	12.5 / 0
input / status	pass	7 0007	7	7 / 0

Address conventions

PDU: 0–65535. One-based: 1–65536, subtract 1. Five-digit references: holding 40001–49999, input 30001–39999; subtract 40001 / 30001. Reference notation is a selected convention, never inferred. Snapshot starts are always PDU addresses.

Byte order

Each recorded word is stored high byte first. For canonical bytes A B C D, map order describes the incoming bytes: ABCD, CDAB, BADC or DCBA. For 16-bit values, AB or BA. Float NaN payloads are preserved in canonical bits; assertions compare NaN category, not payload.

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